$$y_{(x=x_0+nh)} = y_0 + \frac{n}{1!}\Delta y_0 + \frac{n(n-1)}{2!}\Delta^2 y_0 + \frac{n(n-1)(n-2)}{3!}\Delta^3 y_0 + \ldots$$

To find y at $x = 1946$

$\therefore\ x_0 + nh = 1946,\ x_0 = 1941,\ h = 10$

$1941 + n(10) = 1946 \implies n = 0.5$

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
1941	20					
		4				
1951	24		1			
		5		1		
1961	29		2		0	
		7		1		−9
1971	36		3		−9	
		10		−8		
1981	46		−5			
		5				
1991	51					

$$y_{(x=1946)} = 20 + \frac{0.5}{1!}(4) + \frac{0.5(0.5-1)}{2!}(1) + \frac{0.5(0.5-1)(0.5-2)}{3!}(1) + \frac{0.5(0.5-1)(0.5-2)(0.5-3)}{4!}(0) + \frac{0.5(0.5-1)(0.5-2)(0.5-3)(0.5-4)}{5!}(-9)$$

$$= 20 + 2 - 0.125 + 0.0625 - 0.24609$$

$= 21.69$ lakhs

Example 5.18

The following data are taken from the steam table.

Tempreture C^0	140	150	160	170	180
Pressure kg f/cm^2	3.685	4.854	6.302	8.076	10.225

Find the pressure at temperature $t = 175^0$

Solution:

Since the pressure required is at the end of the table, we apply Backward interpolation formula. Let temperature be x and the pressure be y.

$$y_{(x=x_n+nh)} = y_n + \frac{n}{1!}\nabla y_n + \frac{n(n+1)}{2!}\nabla^2 y_n + \frac{n(n+1)(n+2)}{3!}\nabla^3 y_n + \ldots$$

To find y at $x = 175$

$\therefore\ x_n + nh = 175,\ x_n = 180, h = 10 \implies n = -0.5$

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
140	3.685				
		1.169			
150	4.854		0.279		
		1.448		0.047	
160	6.032		0.326		0.002
		1.774		0.049	
170	8.076		0.375		
		2.149			
180	10.225				

$$y_{(x=175)} = 10.225 + (-0.5)(2.149) + \frac{(-0.5)(0-5)}{2!}(0.375) + \frac{(-0.5)(0-5)(1.5)}{3!}(0.049) + \frac{(-0-5)(0.5)(1.5)(2.5)}{4!}(0.002)$$

$$= 10.225 - 1.0745 - 0.046875 - 0.0030625 - 0.000078125$$

$= 9.10048438 = 9.1$

Example 5.19

Calculate the value of y when $x = 7.5$ from the table given below:

x	1	2	3	4	5	6	7	8
y	1	8	27	64	125	216	343	512

Solution:

Since the required value is at the end of the table, apply backward interpolation formula.

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